

WTA CHAMPIONSHIPS

Statistical Analysis

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Why Tennis:

- Dynamic sport, no fixed match length, can last from one hour to over three

- Duration shaped by player attributes and match environment (like surface type)

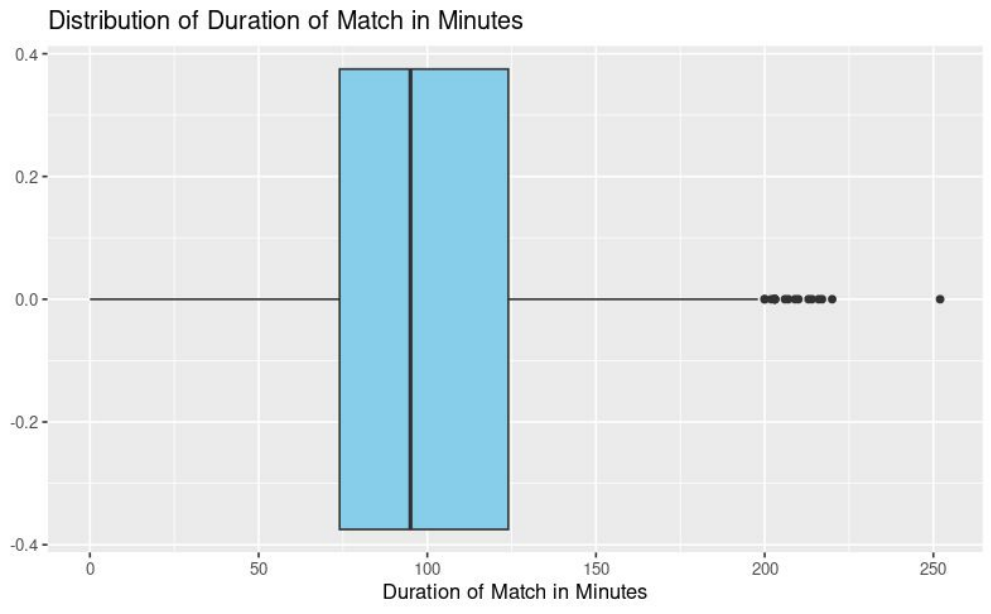
Research Question:

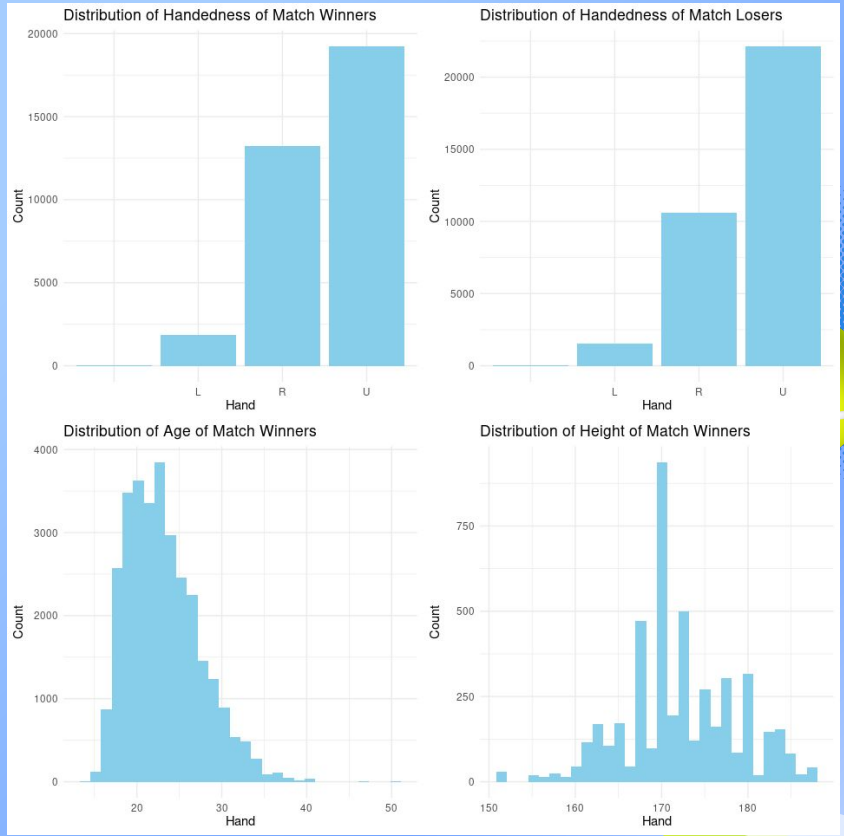
Given court type, age, height, and playing hand of the winner and loser, which factor most influences match duration in women's singles tennis (for the 2023 season)

The Dataset

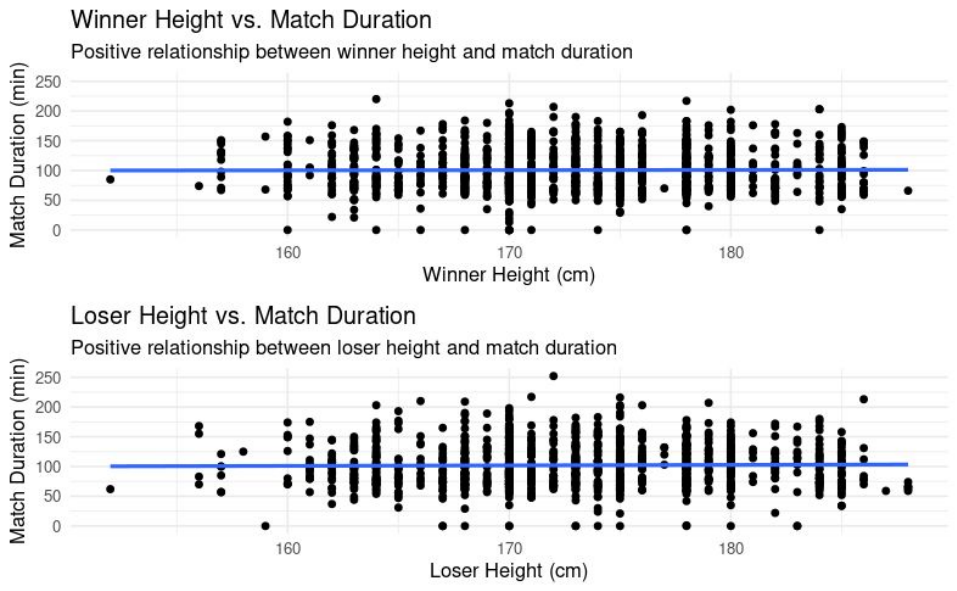
minutes	Duration of the match in minutes - this is our **response variable**
surface	Type of court surface. Categorical: Clay, hard, grass, etc.
winner_hand and loser_hand	Handedness of the winner/loser - R (right), L (left), U (ambidextrous)
winner_age and loser_age	Age of the winner/loser in years
winner_ht and loser_ht	Height of the winner/loser in centimeters
winner_rank and loser_rank	World ranking of the winner/loser at the time of the match

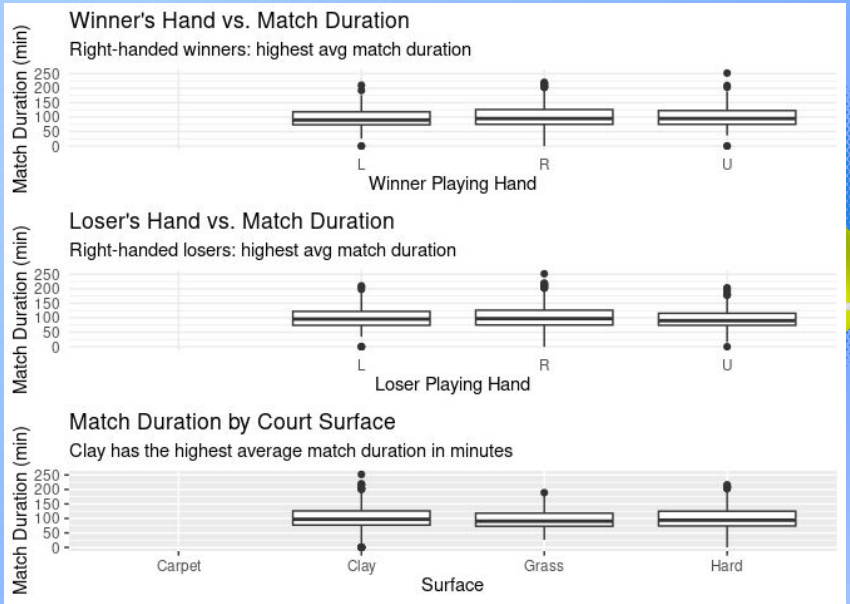
EDA


Distribution of Duration of Match in Minutes
0.4
0.2
0.0
-0.2
-0.4
0
50
100
150
200
250
Duration of Match in Minutes


Distribution of Handedness of Match Winners
Count
20000
15000
10000
5000
0
L
R
U
Hand
Distribution of Handedness of Match Losers
Count
20000
15000
10000
5000
0
L
R
U
Hand
Distribution of Age of Match Winners
Count
4000
3000
2000
1000
0
20
30
40
50
Hand
Distribution of Height of Match Winners
Count
750
500
250
0
150
160
170
180
Hand

EDA


Winner Height vs. Match Duration
Positive relationship between winner height and match duration
Match Duration (min)
250
200
150
100
50
0
160
170
180
Winner Height (cm)
Loser Height vs. Match Duration
Positive relationship between loser height and match duration
Match Duration (min)
250
200
150
100
50
0
160
170
180
Loser Height (cm)


Winner's Hand vs. Match Duration
Right-handed winners: highest avg match duration
Match Duration (min)
250
200
150
100
50
0
L
R
U
Winner Playing Hand
Loser's Hand vs. Match Duration
Right-handed losers: highest avg match duration
Match Duration (min)
250
200
150
100
50
0
L
R
U
Loser Playing Hand
Match Duration by Court Surface
Clay has the highest average match duration in minutes
Match Duration (min)
250
200
150
100
50
0
Carpet
Clay
Grass
Hard
Surface

Results:

A tibble: 3 × 3

Model <chr>	AIC <dbl>	Adj_R2 <dbl>
Surface only	23594.591	0.002
Main effects	6200.058	0.020
With interactions	6204.878	0.025